

Jake Chisholm

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EDUCATION

McMaster University

Hamilton, ON

B.Eng - Computer Engineering — CGPA - 3.9/4.0

Expected Graduation: May 2028

Relevant Coursework

Circuit Analysis & Waves, Software Development, Data Structures & Algorithms, Logic Design, Signals & Systems

WORK EXPERIENCE

Software Engineer Intern

May 2025 - August 2025

Blank Space Corporation

Mississauga, ON

- Developed & tested Node.js backend functions using ES6 features to extract, process & transform JSON data.
- Built multi-page web applications within the Next.js framework, leveraging TypeScript & Tailwind additionally.
- Integrated REST & WebSocket APIs (using AWS AppSync) within web pages; the latter used for a real-time order tracking page.
- Collaborated on a delivery tracking page using Mapbox GL to visualize location & status updates.
- Leveraged the Redux store to implement global & persistent features across codebases.

PROJECTS

DoorSense | *Embedded Systems, Mobile Development, IoT, Edge AI*

October 2025 – Present

- Prototyping a smart door-monitoring device inspired by the Ring camera, using a Raspberry Pi 5 & the Linux-based Raspberry Pi OS.
- Developed a cross-platform React Native mobile app, including live video feed enablement & other device controls.
- Currently developing an IoT pipeline using WebRTC for real-time video streaming & bi-directional communication between the Pi & the app.
- Plans to develop Edge AI algorithms for facial recognition, event inferences & analyzing the occurrence of scheduled events (using the calendar in the app), using OpenCV & TensorFlow Lite.
- Plans to develop a lightweight backend for push notifications from the Pi to the app.

Personal Portfolio Website | *Full-Stack Development*

June 2025 – September 2025

- Developed a personal portfolio website to showcase personal & technical skills more in-depth.
- Programmed in Next.js, making use of vanilla CSS & Tailwind CSS for page styling & TypeScript for type safety.
- Utilized Node.js & nodemailer packages to implement SMTP back-end capabilities for end-user contact.
- Created UI/UX designs on the project, employing the use of Figma to create the designs & user experience for the website.

Game Development Portfolio

- Over the span of many years, using the foundations of OOP principles, I've developed a variety of programming projects.
- Developed the snake game using C++, playable in the cmd terminal, with features using BST's, Hashing & other DSA principles.
- Created a two-player fighting game in Java using the libGDX game engine, featuring multiple characters & menus.
- Developed a re-creation of the NYT Wordle with bonus features incorporated, programmed in Java using JavaFX.
- Created a faithful remake of the arcade game Donkey Kong, programmed in Python using the Pygame engine.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, C#, VBA, HTML/CSS, JavaScript, TypeScript, MATLAB, Simulink, Verilog

Tools & Technologies: LTSpice/PSpice, Oscilloscopes, FPGAs, State Machines, SDLC's & Agile Methodologies, React, Node.js, Jest (Unit Testing), Next.js, React Native, Expo, AWS, Rest APIs, GraphQL, Git, GitHub, Arduino, Raspberry Pi, Figma, UML Diagrams

Hobbies: Bodybuilding, Boxing, Hockey, Guitar, Piano, Music Production (Logic Pro X), Cooking, Reading.